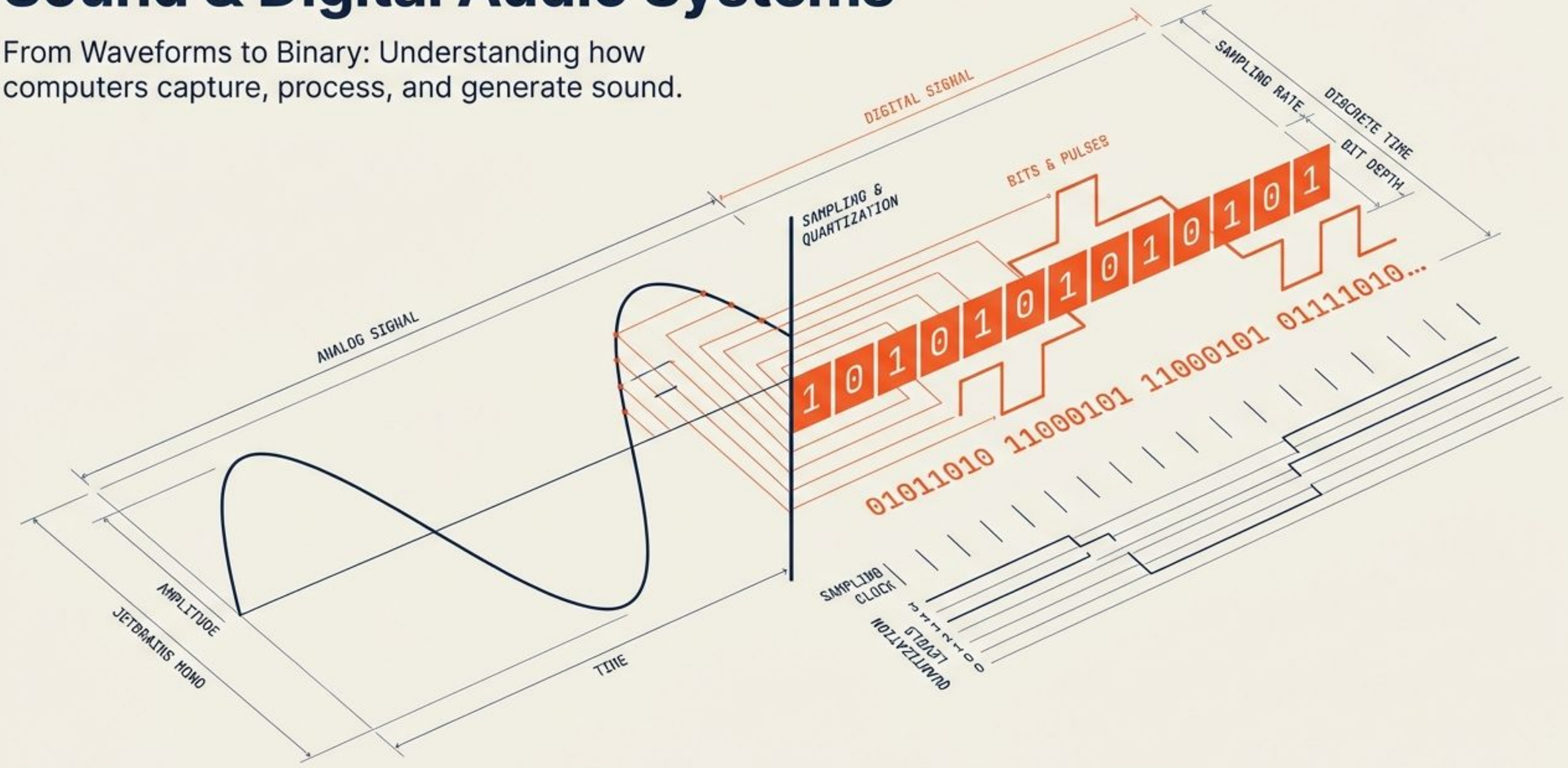
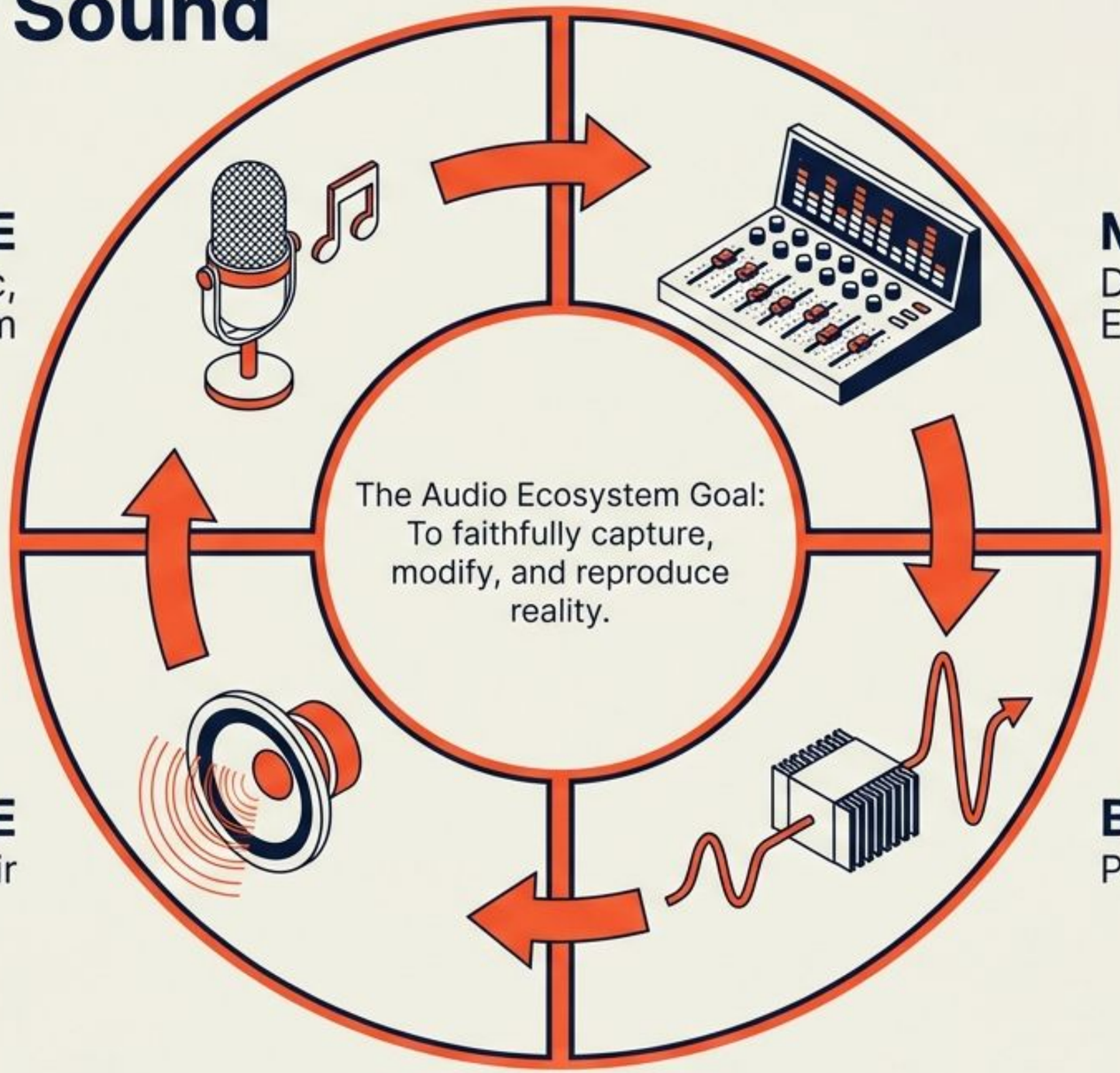
Sound & Digital Audio Systems

From Waveforms to Binary: Understanding how computers capture, process, and generate sound.



Prepared By : Er. Sharat Maharjan

The Cycle of Sound



CAPTURE
Source: Mic, Instrument, Stream

MANIPULATE
DSP, Mixing, Effects

BOOST
Power Amplification

REPRODUCE
Transduction to Air

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Example: Blu-ray (Source) -> AV Receiver (Process/Amp) -> Speakers (Output)
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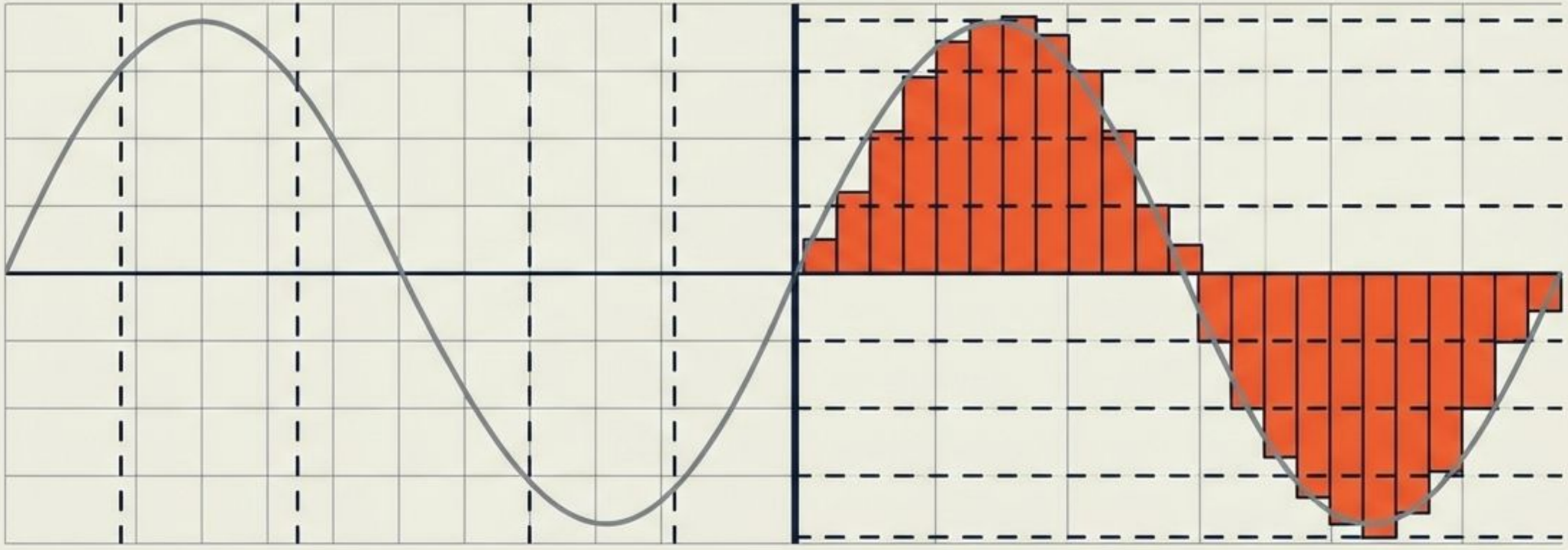
Crossing the Bridge: Analog to Digital

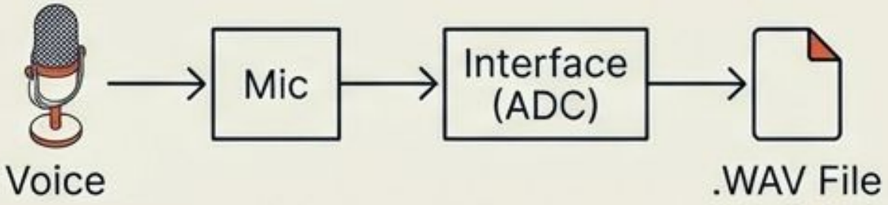
The Reality (Analog Voltage)

The Map (Digital Data)



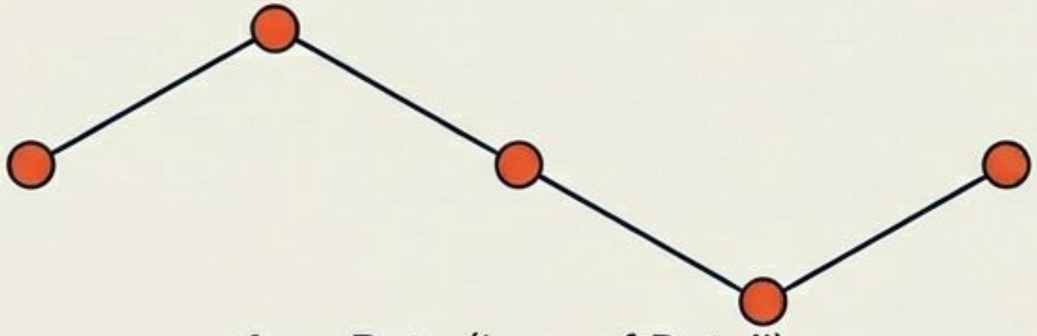
Quantization
(Measuring Height)

Sampling
(Freezing Time)



The Metrics of Digital Fidelity

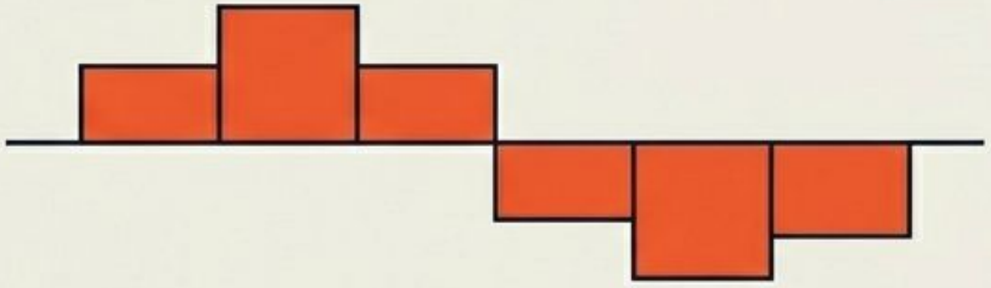
Sampling Rate (Time)



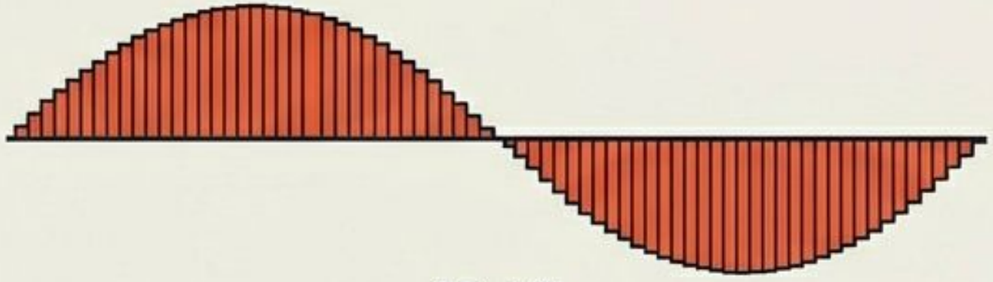
Low Rate (Loss of Detail)

44.1 kHz
(CD Quality)

Bit Depth (Volume/Dynamic Range)



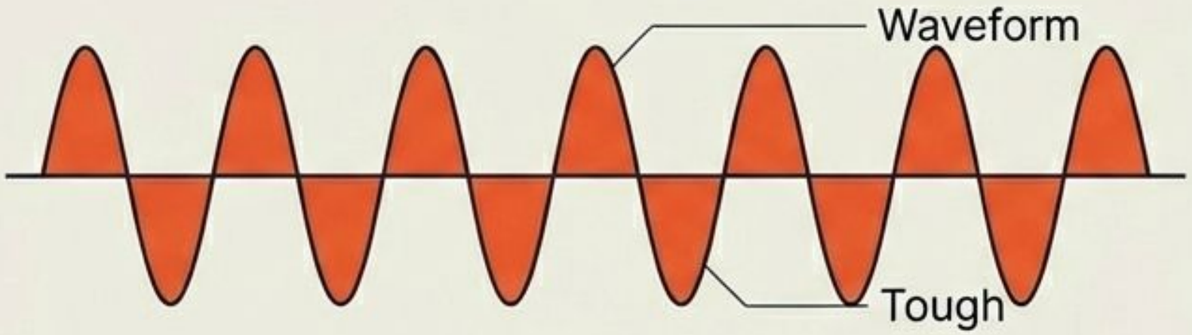
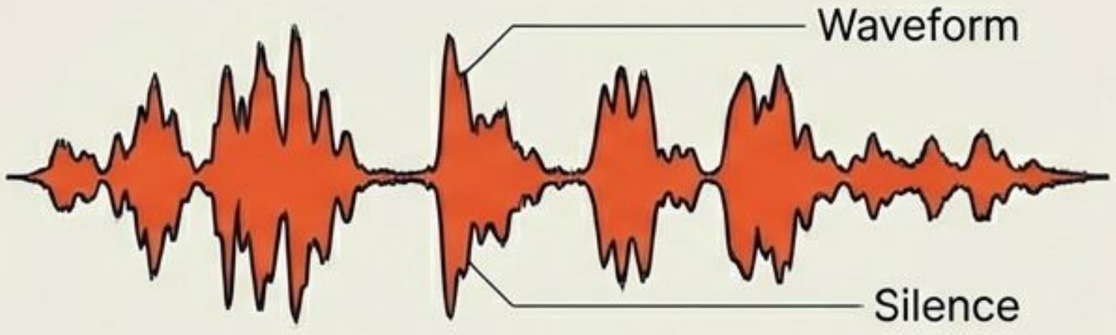
Low Bit Depth (Jagged)



16-bit
(65,536 Levels)

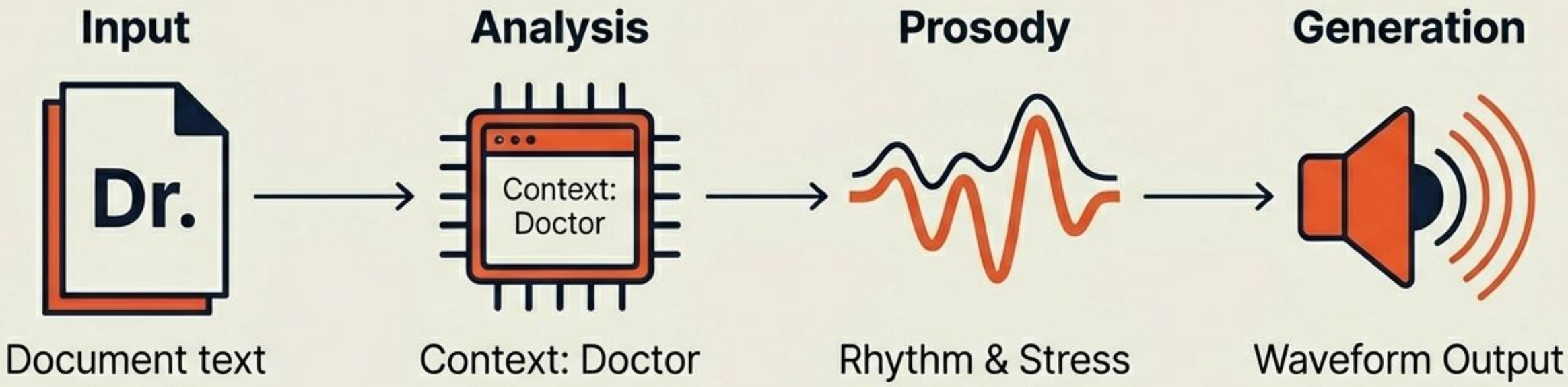
The Golden Standard: 44.1kHz / 16-bit.
This is the mathematical threshold for CD-quality audio.

Distinguishing Signals: Music vs. Speech

Music	Speech
	
Goal: Aesthetic & Emotional Expression	**Goal:** Propositional Meaning & Information
Structure: Discrete Pitches & Isochronic Beat	**Structure:** Continuous contours, Irregular Rhythm
Modulation: Slow, steady changes (1-2 Hz)	**Modulation:** Fast, irregular changes (4-5 Hz)

Music is periodic and emotional. Speech is aperiodic and informational.

Teaching Machines to Speak (TTS)



Old Method (Concatenative)

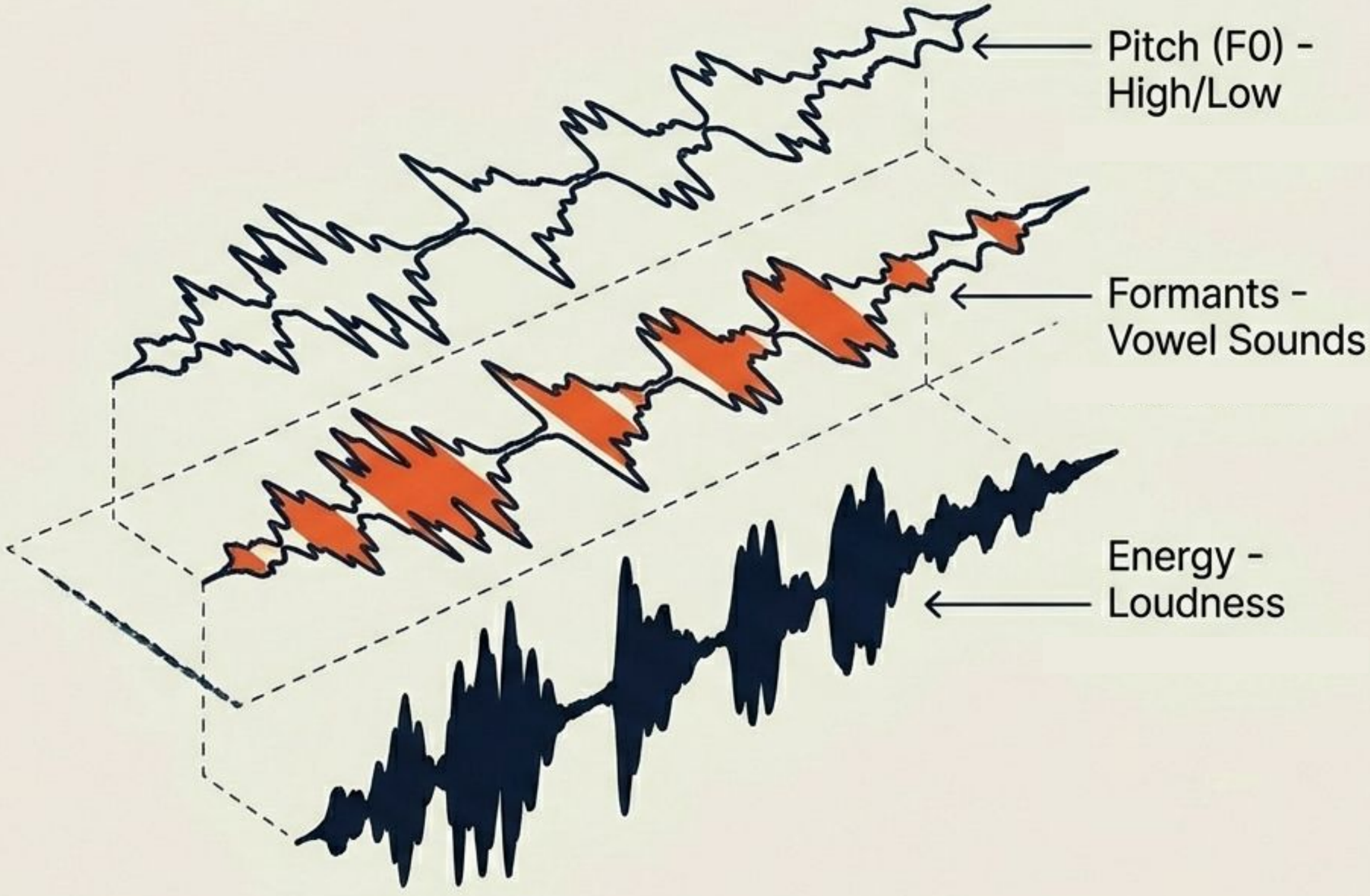
Stitching pre-recorded tape snippets.

New Method (Neural)

Deep learning models generating fresh audio (e.g., Sesame).

Application: Siri, Alexa, Google Assistant.

Teaching Machines to Listen (Analysis)

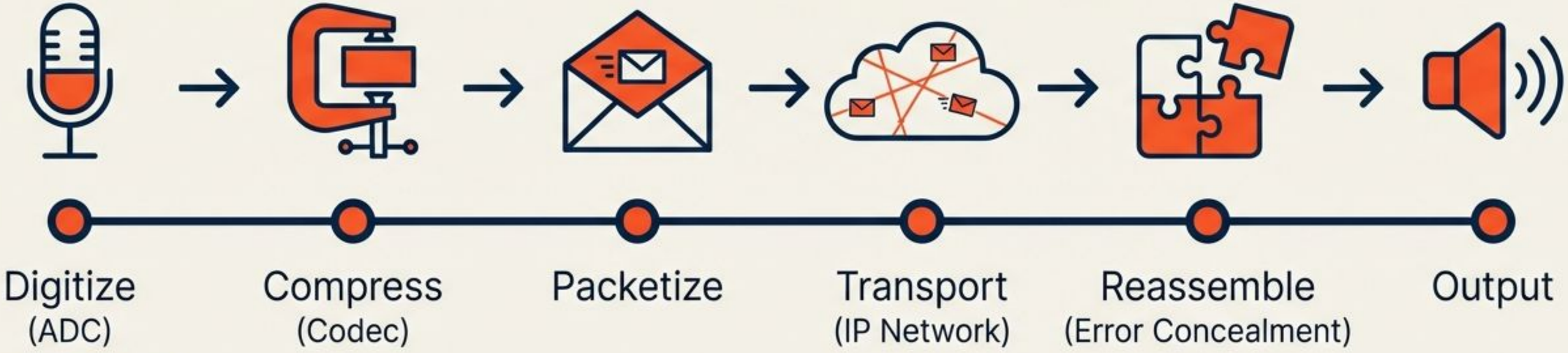


Automatic Speech Recognition (Speech-to-Text)

Speaker Identification (Voice Biometrics)

Trigger: 'Hey Google' - Constant low-level analysis.

The Journey of a Voice (VoIP)



Like putting a letter in an envelope, mailing it through a chaotic system, and taping it back together at the destination.

Preservation Strategies: Audio Formats

Uncompressed

WAV / AIFF

Raw data.
Huge size.
Professional Editing.

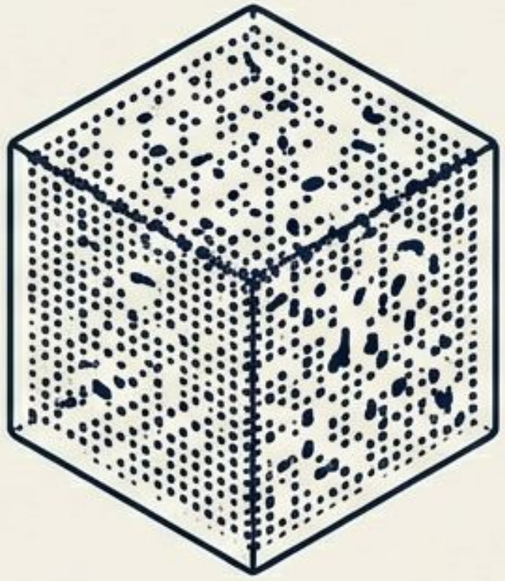
Lossless

FLAC / ALAC

Zipped tight.
Perfect quality.
Audiophile Archive.

Lossy

MP3 / AAC



Perceptual coding.
Discards invisible sound.
Streaming.

MIDI: The Sheet Music, Not the Sound

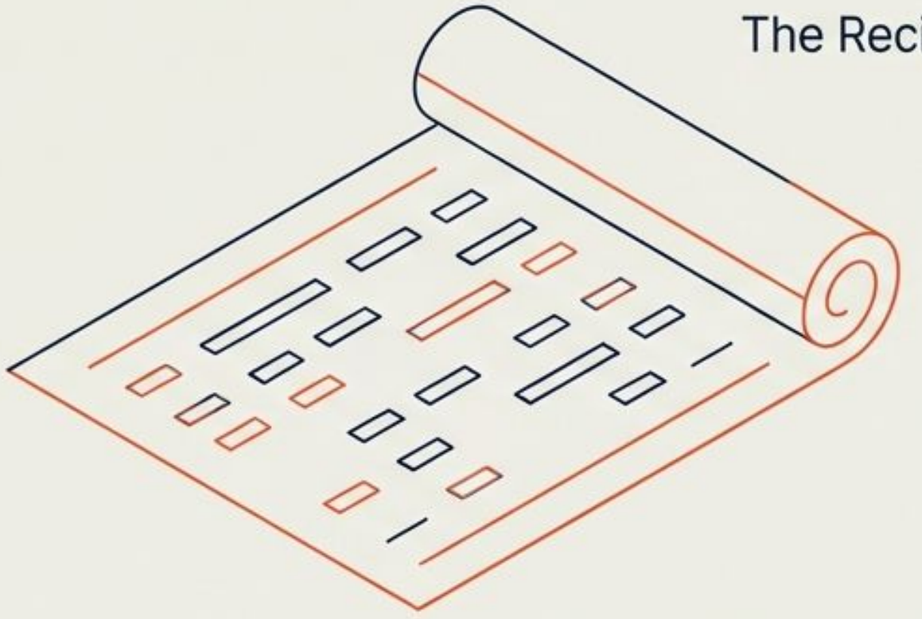
[NOTE ON] | [PITCH: C3] | [VELOCITY: 100] | [CHANNEL: 1]

MIDI contains NO sound.

Musical Instrument Digital Interface is a protocol of commands.
It tells a synthesizer what to play, not what it sounds like.

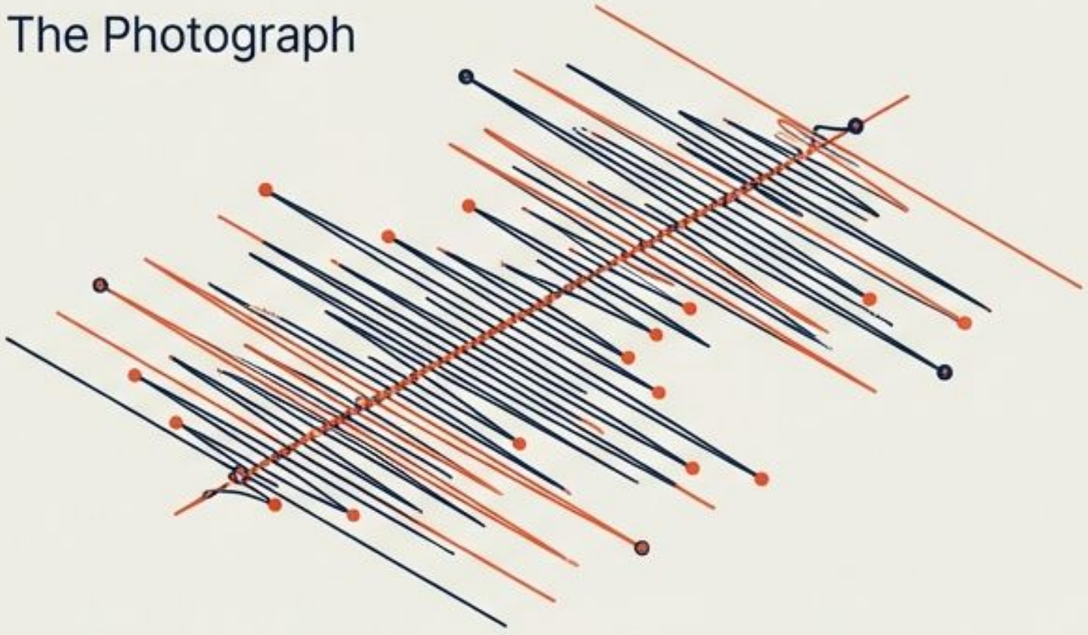
The Showdown: MIDI vs. Digital Audio

The Recipe



- File Size: Tiny (KB)
- Editability: High (Change notes easily)
- Nature: Instructions

The Photograph



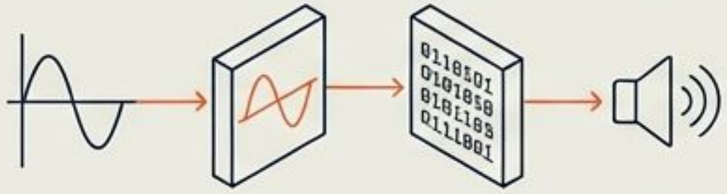
- File Size: Large (MB)
- Editability: Low (Fixed recording)
- Nature: Actual Sound

Use MIDI for composing and ringtones. Use Audio for final music distribution.

System Architecture Recap

ADC / DAC

The bridge between analog reality and digital data.

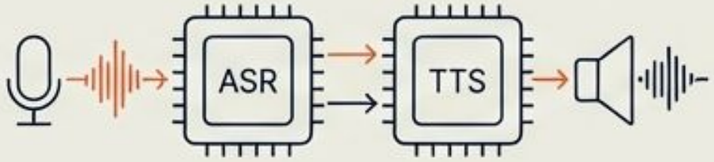


44.1kHz / 16-bit

The CD-Quality standard.
Time accuracy vs. Dynamic range.

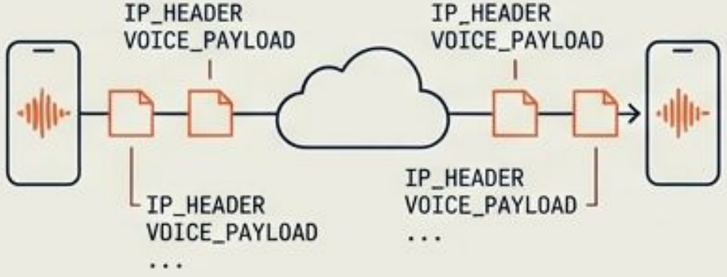
TTS & ASR

Synthesis (Speaking) and Recognition (Listening).



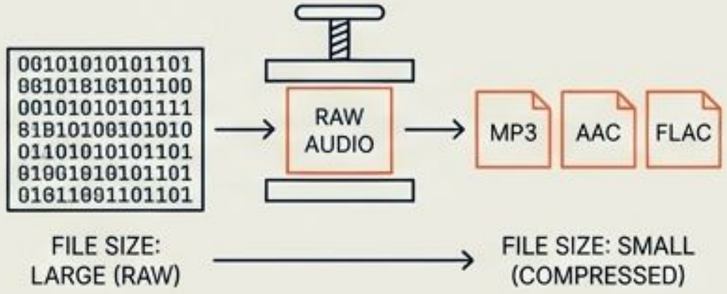
VoIP

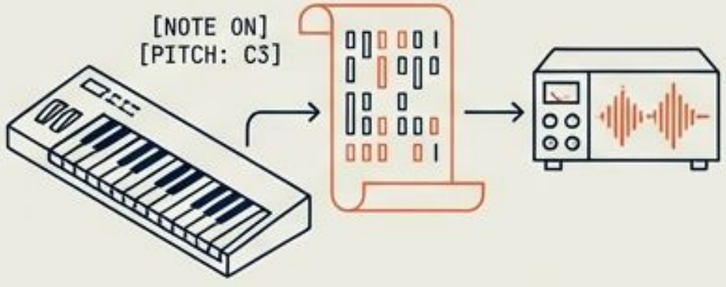
Voice over IP. Packetized transmission of speech.



Codecs

Compression tools (MP3, AAC, FLAC) to manage file size.



MIDI

Digital performance instructions. Not audio files.